Rational Dynamics Notes

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Notation

- $\mathbb{C}_\infty$: Extended complex plane.
- Δ: Unit disk in $\mathbb{C}$.
- N: Neighborhood of a point.
- fg: Composition of f and g ($f \circ g$).
- f^n: The composition of f with itself n times.
- $I = f^0$: Identity function.
- $f^{(n)}$: n-th derivative of f.
- ζ: Fixed point.
- $R(z)$: Rational function in complex variable z.
- σ: Chordal metric.
- σ_0: Spheric metric.
- γ: A curve.
- $E(\gamma)$: Exterior component of a closed curve.
- Ω: An interior component of a closed curve.
- $\nu_f(z_0)$: Valency of f at z_0.
- $m(R, z_0)$: Multiplier of R at z_0.

Prerequisites

From Complex Analysis

- d-fold map: We say f is a d-fold map of V onto W if, for every $w \in W$, the equation $f(z) = w$ has exactly d solutions in V counting multiple solutions (by their multiplicity).
- Maximum Modulus Theorem.
- Schwarz's Lemma.
- Rouché's Theorem.
- Hurwitz's Theorem: If a sequence of univalent analytic maps f_n on a domain D converges uniformly to f, then f is either constant or univalent in D.

From Topology

- Metric spaces: Uniform continuity.
- Compactness.
- Connectedness.

Contents

Chapter 1

Examples

1.1 Introduction

Proposition 1.1 ($z_n \xrightarrow[n\to\infty]{} w \implies w$ is a fixed point)**.** *Let $z_0 \in \mathbb{C}_\infty$, R be a rational function in $\mathbb{C}_\infty$, and $z_n = R^n(z_0)$. Then if z_n converges when n goes to infinity, $\lim_{n\to\infty} z_n = w$ is a fixed point of R.*

Proof. Since R is continuous (everywhere where it is defined, i.e., the denominator is not zero) because it is a rational function, the following is true:

$$w = \lim_{n\to\infty} z_n = \lim_{n\to\infty} R(z_{n-1}) = R\left(\lim_{n\to\infty} z_{n-1}\right) = R(w).$$

Then, $w = R(w)$, so it is a fixed point. □

Intuition: Let's see why the derivative can work for determining if a point is attracting or repelling. Since $|R(z) - \zeta| = |R(z) - R(\zeta)| \simeq |R'(\zeta)||z - \zeta|$ if z is close enough to ζ, we have that if $|R'(\zeta)| < 1$, $z_n \to \zeta$, and if $|R'(\zeta)| > 1$, z_n will be initially repelled from ζ. (A rigorous contradiction proof follows in the notes ensuring z_{n+1} cannot arbitrarily stay closer than z_n if repelled).

1.2 Iteration of Möbius Transformations

Definition 1.1 (Möbius Transformation)**.** A *Möbius transformation* is a rational function $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$ of the form

$$R(z) = \frac{az + b}{cz + d},$$

where $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$.

Theorem 1.2 (Classification of Möbius Transformations)**.** *Given* R *a Möbius transformation, then either it has:*

1. *One fixed point,* ζ*, and for every* $z \in \mathbb{C}$*,* $R^n(z) \to \zeta$*.*
2. *Two fixed points,* ζ_1 *and* ζ_2*, and for every* $z \in \mathbb{C}_\infty$ *either:*
   (a) $R^n(z) \to \zeta_1$
   (b) $R^n(z) \to \zeta_2$
   (c) $R^n(z)$ *moves cyclically over a finite set of points.*
   (d) $\{R^n(z) : n \in \mathbb{N}_0\}$ *forms a dense subset of a circle.*

Proof. First, observe that R cannot have 0 fixed points. Solving $R(z) = z$:

$$\frac{az+b}{cz+d} = z \implies cz^2 + (d-a)z - b = 0$$

This is a quadratic equation, so it always has 1 or 2 solutions in $\mathbb{C}_\infty$ (if $c = 0$, ∞ is a fixed point).

Case 1: One fixed point. Suppose the fixed point is ∞. Then $R(z) = z + \beta$ for some $\beta \in \mathbb{C}$:

$$R(\infty) = \infty \implies c = 0 \implies R(z) = \frac{a}{d}z + \frac{b}{d} = \alpha z + \beta.$$

We also want R to have no other fixed point:

$$R(z) = z \iff \alpha z + \beta = z \iff (\alpha - 1)z + \beta = 0,$$

hence β has to be $\neq 0$ (otherwise $z = 0$ is a fixed point) and $\alpha = 1$ (otherwise $z = \beta/(\alpha - 1)$ is a fixed point).

If R has a general fixed point ζ, we can define the Möbius transformation S by using the conjugation $g(z) = \frac{1}{z-\zeta}$ and $S = g \circ R \circ g^{-1}$. Since conjugation preserves the number of fixed points, S has exactly one fixed point at ∞, meaning S is a translation $S(z) = z + \beta$. Thus, $S^n(z) \to \infty$, which implies $R^n(z) \to \zeta$.

Case 2: Two fixed points. Suppose the fixed points are 0 and ∞ (it can be shown similarly to the previous case). Then $R(z) = kz$ for some $k \in \mathbb{C} \setminus \{0, 1\}$. Analyzing $R^n(z) = k^n z$:

- If $|k| < 1$, then $R^n(z) \to 0$.
- If $|k| > 1$, then $R^n(z) \to \infty$.

- If $|k| = 1$ and k is a root of unity, it cycles over a finite set of points.
- If $|k| = 1$ and k is not a root of unity, the subset will be dense in a circle $|z| = |z_0|$.

For a general map with two fixed points ζ_1, ζ_2, we conjugate via $g(z) = \frac{z-\zeta_1}{z-\zeta_2}$ to place the fixed points at 0 and ∞, yielding the same behavioral classification. $\square$

1.3 Iteration of $z \mapsto z^2$

Definition 1.2 (Forward and Backward Invariant)**.** Given a set $C \subseteq \mathbb{C}_\infty$ and a rational function $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$, we say C is:

1. *Forward invariant* if $R(C) \subseteq C$.
2. *Backward invariant* if $R^{-1}(C) \subseteq C$.

Definition 1.3 (Periodic Point)**.** Given a rational function $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$ and $z \in \mathbb{C}_\infty$, we say z is *periodic with period* k if $R^k(z) = z$.

In the Julia set of $z \mapsto z^2$ (which is $J = \{z : |z| = 1\}$), two close points may behave radically differently. And it has this properties:

1. Given I an arc in J, $\exists N \in \mathbb{N}_0$ such that $\forall n \geqslant N$, $R^n(I) = J$.
2. $\exists z_0 \in J$ such that $\{R^n(z_0) \mid n \in \mathbb{N}_0\}$ is dense in J.
3. $\exists z_0 \in J$ such that $\{R^n(z_0) \mid n \in \mathbb{N}_0\}$ is finite.

1.4 Iteration of $z \mapsto z^2 - 1$

For polynomials like $P(z) = z^2 - 1$, points outside a certain radius escape to infinity ($z \in F_\infty \implies P^n(z) \to \infty$).

$P(F_0) = F_{-1}$, $P(F_{-1}) = F_0$.

$P(\Omega) \to \ldots, F_0, F_{-1}, F_0, \ldots$, Ω a Fatou set $\neq F_\infty$.

1.5 Iteration of $z \mapsto z^2 + c$

Recall Vieta's formulas. Given a polynomial $P(x) = ax^2 + bx + c$ with roots r_1, r_2:

$$r_1 + r_2 = -\frac{b}{a}, \quad r_1 r_2 = \frac{c}{a}.$$

In general, given a polynomial of degree n, $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$

$$\sum_{i=0}^{n} r_i = -\frac{a_{n-1}}{a_n}, \quad \prod_{i=0}^{n} r_i = (-1)^n \frac{a_0}{a_n},$$

and

$$\sum_{0 \leqslant i < j \leqslant n}^{n} r_i r_j = \frac{a_{n-2}}{a_n}, \quad \sum_{0 \leqslant i < j < k \leqslant n}^{n} r_i r_j r_k = \frac{a_{n-3}}{a_n}, \quad \cdots$$

A second degree polynomial can be transformed into a composition of two affine transformations and z^2:

$$P(z) = z - z^2 = -(z^2 - z) = -\left(z^2 - z + \frac{1}{4} - \frac{1}{4}\right) = -\left(\left(z - \frac{1}{2}\right)^2 - \frac{1}{4}\right).$$

Then defining

$$h(z) = z - \frac{1}{2} \quad , \quad g(z) = z^2 \quad , \quad f(z) = -\left(z - \frac{1}{4}\right) = \frac{1}{4} - z,$$

we have

$$P(z) = f \circ g \circ h(z) = f \circ g\left(z - \frac{1}{2}\right) = f\left(\left(z - \frac{1}{2}\right)^2\right) = -\left(\left(z - \frac{1}{2}\right)^2 - \frac{1}{4}\right) = -(z^2 - z).$$

1.6 Iteration of $z \mapsto z + 1/z$

We consider the dynamics of the rational map defined by:

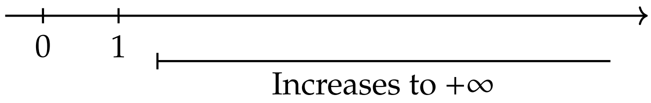
$$R(z) = z + \frac{1}{z}$$

Real Axis:

For real values $z = x \in \mathbb{R}$ with $x > 1$:

$$R(x) = x + \frac{1}{x} > x.$$

The sequence of iterates increases monotonically towards $+\infty$.

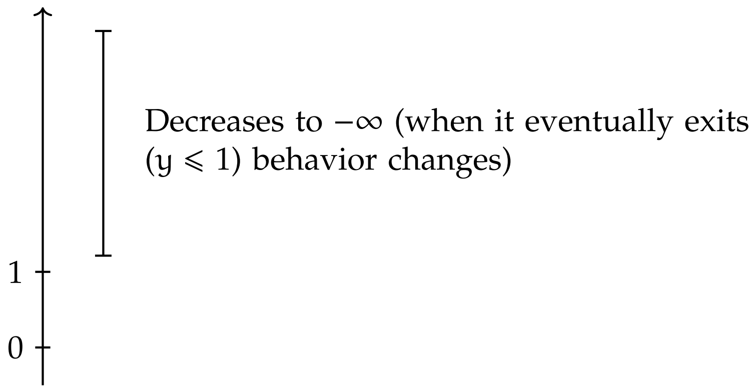


Imaginary Axis:

For purely imaginary values $z = iy \in i\mathbb{R}$ with $y > 1$:

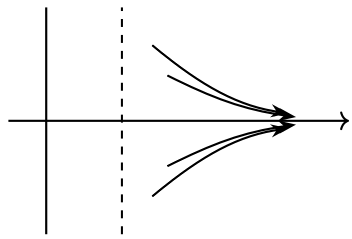
$$R(iy) = iy + \frac{1}{iy} = i\left(y - \frac{1}{y}\right).$$

Since $y > 1$, $y - \frac{1}{y} < y$. Thus, the imaginary part decreases towards 0 as long as $y > 1$. Once it eventually exits this regime ($y \leqslant 1$), the behavior changes completely.



Region $x > 1$:

In the half-plane where $\mathrm{Re}(z) > 1$, the trajectories flow outwards towards the right along curves that flatten asymptotically.



Analyzing the Conjugate Map:

We can understand the behavior at infinity by conjugating via $g(z) = \frac{1}{z}$, which maps the neighborhood of ∞ to a neighborhood of the origin. Let $S = g \circ R \circ g^{-1}$. Since $g^{-1}(z) = \frac{1}{z}$, we have:

$$S(z) = g\left(R\left(\frac{1}{z}\right)\right) = \frac{1}{\frac{1}{z} + z} = \frac{z}{1+z^2},$$

where $g = \frac{1}{z}$. See figure 1.1 for the action of R near ∞.

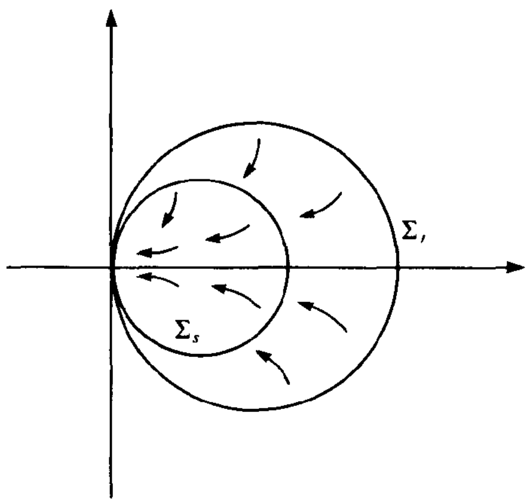


Figure 1.1: Action of $z \mapsto z + 1/z$ near ∞.

1.7 Iteration of $z \mapsto 2z - 1/z$

For $R(z) = \frac{1}{2}(z - \frac{1}{z})$, we find the fixed points:

- $\zeta_1 = 1$ (repelling),

- $\zeta_2 = -1$ (repelling),
- $\zeta_3 = \infty$ (attracting).

1.8 Newton's Approximation

To find the roots of a polynomial function f in $\mathbb{R}$, the Newton-Raphson method uses the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Given f in $\mathbb{C}$, we can define the rational function:

$$R(z) = z - \frac{f(z)}{f'(z)}.$$

And the analysis of these kinds of functions (rational functions), falls within the scope of this text.

Note that, in $\mathbb{R}$ that kind of approximation works, but in $\mathbb{C}_\infty \ldots$ the conditions of z_0 to converge to a root are hard and have to do with the Julia set of R (which for polynomials f with degree 3 or more is usually a complex fractal).

Properties of the Newton method:

1. The roots of f correspond with fixed points of R (proved in the book [1]).
2. Successive iterates of R have a tendency to converge to an attracting fixed point of R.

Thus, if ζ is a zero of f which gives rise to an attracting fixed point of F, then,

$$z_n = F^n(z_0) \xrightarrow[n\to\infty]{} \zeta$$

for at least initial guesses z_0 of ζ sufficiently accurate.

When z_0 is far from ζ the problem of deciding if $z_n \to \zeta$ is much harder.

In the book ([1]) there is analysis of polynomials of degree 2 with distinct roots.

A formulation of a theorem for polynomial with real distinct roots: z_n converges to a root of P for essentially every z_0 (except from a set with area 0)

1.9 General Remarks

For typical rational functions R, the extended complex plane $\mathbb{C}_\infty$ is divided into J (Julia set) and F (Fatou set).

- F preserves proximity.
- J is in every sense chaotic.
- J and F are both forward and backward invariant.

Chapter 2

Rational Maps

2.1 The Extended Complex Plane

Definition 2.1 (Extended Complex Plane)**.** The *extended complex* plane is the union

$$\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\},$$

where ∞ is an "abstract" point outside $\mathbb{C}$.

Now let's define a metric in $\mathbb{C}_\infty$ that handles all points with equal ease and makes ∞ lose any special significance: *the chordal metric*.

Let's start by defining the *Riemann Sphere*. For that, we identify $\mathbb{C}$ with the horizontal plane

$$\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_3 = 0\}$$

in $\mathbb{R}^3$. Let S be the unit sphere in $\mathbb{R}^3$ centered at the origin, and $N = (0, 0, 1)$ be the north pole.

We now project each point in $\mathbb{C}$ to a point in S by taking the intersection between the line that crosses the point $(x_1, x_2, 0)$ and N with S. An illustration of this projection is given in Figure 2.1.

We identify this projection as:

$$\pi : \mathbb{C}_\infty \longrightarrow S,$$

where for $z = x + iy \in \mathbb{C}$:

$$\pi(z) = \left(\frac{2x}{1 + |z|^2}, \frac{2y}{1 + |z|^2}, \frac{|z|^2 - 1}{1 + |z|^2}\right),$$

and $\pi(\infty) = (0, 0, 1)$.

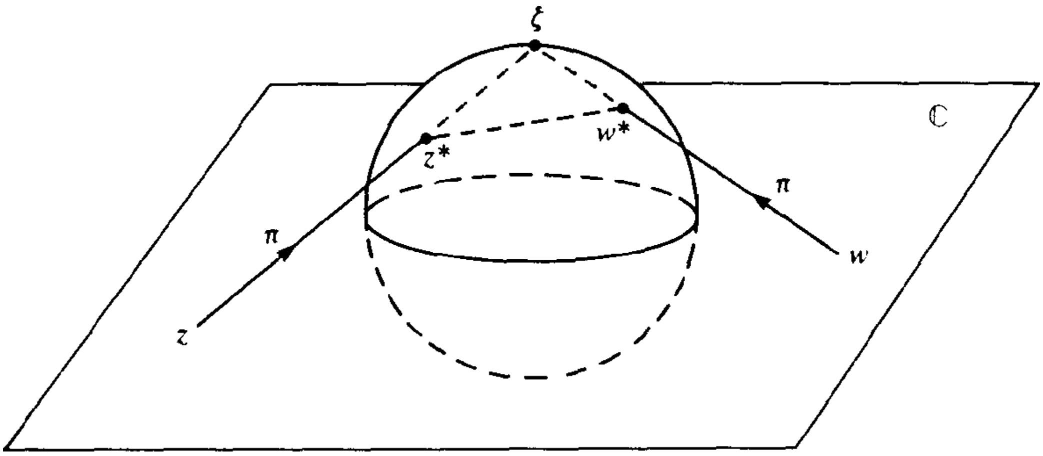


Figure 2.1: Stereographic projection.

Definition 2.2 (Chordal Metric)**.** Given the projection π from $\mathbb{C}_\infty$ to S, we define the *chordal metric*, $\sigma : \mathbb{C}_\infty \times \mathbb{C}_\infty \to \mathbb{R}^+ \cup \{0\}$, as the Euclidean distance from $\pi(z)$ to $\pi(w)$:

$$(z, w) \mapsto \sigma(z, w) = |\pi(z) - \pi(w)|.$$

Explicitly, when $z, w \in \mathbb{C}$:

$$\sigma(z, w) = \frac{2|z - w|}{(1 + |z|^2)^{1/2}(1 + |w|^2)^{1/2}}.$$

For the point at infinity, we have:

$$\sigma(z, \infty) = \lim_{w \to \infty} \sigma(z, w) = \frac{2}{(1 + |z|^2)^{1/2}}.$$

Remark 2.1 ($z \mapsto 1/z$ is an isometry). $\sigma(z, w) = \sigma(z^{-1}, w^{-1})$, i.e., the map $z \mapsto \frac{1}{z}$ is an isometry.

Remark 2.2 (Rotation in $S \mapsto$ isometry in $\mathbb{C}_\infty$). Any rotation $\varphi : S \to S$ of the sphere induces a map $\pi^{-1}\varphi\pi : \mathbb{C}_\infty \to \mathbb{C}_\infty$ which is an isometry. These isometries are Möbius transformations of the form:

$$z \mapsto \frac{az - \bar{c}}{cz + \bar{a}},$$

where $|a|^2 + |c|^2 = 1$.

The chordal metric handles the point at infinity with ease, but it measures distance through the three-dimensional space containing the Riemann sphere. If we restrict our measurements strictly to the two-dimensional curved surface of the sphere, we obtain the spherical metric.

Definition 2.3 (Spherical Metric)**.** Given the stereographic projection $\pi : \mathbb{C}_\infty \to S$, the *spherical distance* $\sigma_0(z, w)$ between two points in $\mathbb{C}_\infty$ is defined as the Euclidean length of the shortest path (an arc of a great circle) connecting their projections $\pi(z)$ and $\pi(w)$ on the surface of S.

If the chord joining the projected points subtends an angle θ at the origin of the sphere, then by definition the spherical distance is exactly $\sigma_0(z, w) = \theta$. By using the trigonometric connection to the chordal metric, we can derive explicit formulas directly in terms of the complex coordinates for $z, w \in \mathbb{C}$:

$$\sigma(z, w) = 2 \sin\left(\frac{\theta}{2}\right) = 2 \sin\left(\frac{\sigma_0(z, w)}{2}\right).$$

Thus,

$$\begin{aligned}\sigma_0(z, w) &= 2 \arcsin\left(\frac{\sigma(z, w)}{2}\right) \\ &= 2 \arcsin\left(\frac{|z - w|}{(1 + |z|^2)^{1/2}(1 + |w|^2)^{1/2}}\right).\end{aligned}$$

For the point at infinity, taking the limit as $w \to \infty$ yields:

$$\sigma_0(z, \infty) = 2 \arcsin\left(\frac{1}{(1 + |z|^2)^{1/2}}\right).$$

Remark 2.3 ($\sigma \sim \sigma_0$). The chordal and spherical metrics are topologically equivalent ($\sigma(z, w) = 2 \sin(\theta/2)$). Furthermore, they satisfy the following inequalities:

$$\frac{2}{\pi}\sigma_0(z, w) \leqslant \sigma(z, w) \leqslant \sigma_0(z, w)$$

Proof. The inequalities follow directly from applying the elementary real-variable inequality $\frac{2\theta}{\pi} \leqslant \sin\theta \leqslant \theta$ on the interval $0 \leqslant \theta \leqslant \frac{\pi}{2}$ to the angle $\theta = \frac{\sigma_0(z,w)}{2}$. □

Because these metrics differ by at most a bounded scale factor of $\frac{2}{\pi}$, we can use σ and σ_0 interchangeably throughout our study of rational dynamics whenever global constants are allowed to change.

Remark 2.4 (Spherical length). Given a continuously differentiable curve $\gamma : I \to \mathbb{C}_\infty$, its *spherical length* is given by:

$$\int_\gamma \frac{2|dz|}{1 + |z|^2}.$$

2.2 Rational Maps

Definition 2.4 (Rational Map)**.** A *rational map* is a function of the form

$$R(z) = \frac{P(z)}{Q(z)},$$

where $P(z)$ and $Q(z)$ are polynomials, not both being the zero polynomial.

- If $P = 0 \Rightarrow R = 0$.
- If $Q = 0 \Rightarrow R = \infty$.
- If $P(z_0) \neq 0$ and $Q(z_0) = 0$ for some $z_0 \in \mathbb{C}$, then $R(z_0) = \infty$.
- We define $R(\infty) = \lim_{z\to\infty} R(z)$.
- If $P(z_0) = 0$ and $Q(z_0) = 0$, strictly speaking $R(z_0)$ is not defined, but in this case we may cancel the common factors:

$$P(z) = (z - z_0)^m P_1(z)$$
$$Q(z) = (z - z_0)^n Q_1(z)$$

where $P_1(z_0) \neq 0$ and $Q_1(z_0) \neq 0$. Then:

$$R(z) = \frac{P_1(z)}{Q_1(z)}(z - z_0)^{m-n}.$$

From now on, we will consider that this has been done, i.e., P and Q are coprime. Thus, given R, the polynomials P and Q are uniquely determined up to scalar multiplication. This allows us to define the degree of a rational map.

Definition 2.5 (Degree of a Rational Map, $\deg(R)$)**.** Given a rational map $R = P/Q$ where P and Q are coprime polynomials, the *degree of* R is defined as:

$$\deg(R) = \max\{\deg(P), \deg(Q)\}$$

where $\deg(\cdot)$ is the usual degree of a polynomial. We will consider that if $R = \alpha \in \mathbb{C}_\infty$, then $\deg(R) = 0$.

Proposition 2.5 ($\deg(RS) = \deg(R)\deg(S)$)**.** *Given two rational function* R, S, *the degree of* RS *is* $\deg(R)$ *times* $\deg(S)$. *That is,*

$$\deg(RS) = \deg(R)\deg(S).$$

Remember that by writing RS *we mean composition* ($R \circ S$) *not multiplication. However,* $\deg(R)\deg(S)$ *is a multiplication.*

Proof. See [1, p. 32]. □

Corollary 2.5.1 ($\deg(R^n) = [\deg(R)]^n$)**.** *Given a rational function* R*, the degree of* R^n *is the* n*-th power of* $\deg(R)$*. That is,*

$$\deg(R^n) = [\deg(R)]^n.$$

Now, let's remember the definitions of *holomorphic*, *meromorphic* and *analytic* functions in $\mathbb{C}$ and see how they translate to $\mathbb{C}_\infty$ with the chordal metric.

Definition 2.6 (Holomorphic)**.** Given $D \subseteq \mathbb{C}$, a function $f : D \to \mathbb{C}$ is *holomorphic* in D if the derivative of f exists at each point in D.

Definition 2.7 (Meromorphic)**.** Given $D \subseteq \mathbb{C}$, a function $f : D \to \mathbb{C}_\infty$ is *meromorphic* in D if each point in D has a neighborhood on which either f or $1/f$ is holomorphic.

Definition 2.8 (Pole of f)**.** Given a meromorphic function f in $D \subseteq \mathbb{C}$, the *poles of* f are the points $w \in D$ where $f(w) = \infty$ ($\lim_{z\to w} f(z) = \infty$).

Remark 2.6 (f meromorphic $\implies$ $1/f$ holomorphic at f's poles). Given a meromorphic function f in $D \subseteq \mathbb{C}$ with a pole at w, then $1/f$ is holomorphic near w and $\frac{1}{f(w)} = 0$.

Definition 2.9 (f is defined near ∞)**.** A function f is said to be *defined near* ∞ (or *in a neighborhood of* ∞) if it is defined on some set $\{z \in \mathbb{C} \mid |z| > r\} \cup \{\infty\}$.

Definition 2.10 (Holomorphic or Meromorphic at ∞)**.** Given a function f defined near ∞, f is *holomorphic* (or *meromorphic*) *at* ∞ if $f(1/z)$ is holomorphic (or meromorphic) near 0.

Definition 2.11 (Analytic in $D \subseteq \mathbb{C}_\infty$)**.** Given a map $f : D_1 \to D_2$ between any two subdomains of $\mathbb{C}_\infty$, we say f is *analytic* if it is holomorphic or meromorphic at each point in D_1.

Remark 2.7 (f meromorphic $\implies$ f continuous at its poles). Given a meromorphic function f in $D \subseteq \mathbb{C}$ with a pole at w, then (using the chordal metric and its topology), f is continuous at the pole. This is because $z \mapsto 1/z$ is a σ-isometry:

$$\sigma(f(z), f(w)) = \sigma\left(\frac{1}{f(z)}, \frac{1}{f(w)}\right) = \sigma\left(\frac{1}{f(z)}, 0\right) = \frac{2|1/f(z)|}{(1 + |1/f(z)|^2)^{1/2}} \xrightarrow{z\to w} 0$$

Summing up, we have that the concepts of *holomorphic*, *meromorphic* and *analytic* functions translate well from $\mathbb{C}$ with the Euclidean metric, to $\mathbb{C}_\infty$ with the chordal or spheric metric. Hence, making ∞ lose any special significance it may have had in $\mathbb{C}$.

Proposition 2.8 (∞ is a fixed point for all polynomials in $\mathbb{C}_\infty$)**.** *Given a polynomial* $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ *where* $n > 0$ *and* $a_n \neq 0$*. Then,* P *is analytic in* $\mathbb{C}_\infty$*.*

Proof. This proof serves to illustrate what holomorphic and meromorphic mean. We have to show that P is holomorphic or meromorphic at each point in $\mathbb{C}_\infty$. We have that P is analytic in $\mathbb{C}$ because it's a polynomial, thus, we only have to show that P is holomorphic or meromorphic at ∞. We'll see that it's meromorphic at ∞, i.e., $P(1/z)$ is meromorphic at 0, i.e., $1/P(1/z)$ is holomorphic at 0. And we can see that

$$\frac{1}{P(1/z)} = \frac{z^n}{a_0 z^n + \cdots + a_n}$$

is differentiable at 0, therefore, P is meromorphic at ∞. □

Theorem 2.9 (Analytic in $\mathbb{C}_\infty \iff$ rational map)**.** *The rational maps are precisely the analytic maps of* $\mathbb{C}_\infty$ *into itself.*

Theorem 2.10 (Rational maps are d-fold maps in $\mathbb{C}_\infty$)**.** *Given* R *a rational function of positive degree* d*. Then* R *is a* d*-fold map of* $\mathbb{C}_\infty$ onto *itself: that is, for any* w *in* $\mathbb{C}_\infty$*, the equation* $R(z) = w$ *has* precisely d *solutions in z (counting multiplicities).*

Proof. See [1, pp. 31–32]. □

Remark 2.11 (Zeros and poles of rational maps). Given $R = P/Q$ a rational function of positive degree d, where P and Q are coprime polynomials with $\deg(P) = n$ and $\deg(Q) = m$. Then, the zeros of R are the zeros of P, and the poles of R are the zeros of Q. Moreover, the number of zeros and poles of R at ∞ (counting multiplicities) are given by the following cases:

1. if $n = m$, then all the zeros and poles of R are in $\mathbb{C}$;
2. if $n > m$, then the number of poles at ∞ is $n - m$;
3. and if $n < m$, then the number of zeros at ∞ is $m - n$.

2.3 The Lipschitz Condition

Theorem 2.12 (Rational maps satisfy the Lipschitz condition)**.** *A rational map* R *satisfies some* Lipschitz condition

$$\sigma_0(R(z), R(w)) \leqslant M\sigma_0(z, w)$$

on the complex sphere.

Proof. See [1, p. 33]. □

Theorem 2.13 (Möbius transformations explicit Lipschitz condition)**.** *The Möbius map*

$$g(z) = \frac{az + b}{cz + d}, \quad ad - bc = 1,$$

satisfies the Lipschitz condition

$$\sigma_0(g(z), g(w)) \leqslant \|g\|^2 \sigma_0(z, w).$$

Proof. See [1, pp. 33–34]. □

Theorem 2.14 (Möbius transformations *family* explicit uniform Lipschitz condition)**.** *Let m be any positive number. Then the Möbius transformations* g *which satisfy*

$$\begin{aligned} \sigma(g(0),\ g(1)) &\geqslant m, \\ \sigma(g(1),\ g(\infty)) &\geqslant m, \\ \sigma(g(\infty), g(0)) &\geqslant m \end{aligned}$$

also satisfy the uniform Lipschitz condition

$$\sigma(g(z), g(w)) \leqslant \frac{\pi}{m^3} \sigma(z, w).$$

Proof. See [1, pp. 34–35]. □

Definition 2.12 (Complementary components of a closed curve γ)**.** Let γ be a *closed curve* on the sphere. The complement of γ is a union of mutually disjoint domains, which we call the *complementary components* of γ.

Definition 2.13 (Exterior and interior of a closed curve γ contained in a open hemisphere)**.** Let γ be a *closed curve* that lies in some *open hemisphere.* Then exactly one of the complementary components contains a hemisphere; we denote this by $E(\gamma)$ and call it the *exterior* of γ. The other complementary components of γ are said to be the *interior components* of γ.

Theorem 2.15 (Interior components mapping theorem for rational maps)**.** *Let* R *be a rational function. Then there is a positive number* δ *such that if* γ *is any closed curve of* σ_0*-diameter less than* δ*, then the image* $\mathsf{R}(\Omega)$ *of an interior component* Ω *of* γ *does not meet the exterior* $\mathsf{E}(\mathsf{R}(\gamma))$ *of* $\mathsf{R}(\gamma)$*.*

Proof. See [1, pp. 35–36]. □

This result almost says that if the curve γ is sufficiently small, then the interior components of γ map to the interior components of the image curve $\mathsf{R}(\gamma)$: however, the images of the interior components may also contain points of $\mathsf{R}(\gamma)$, so it is necessary to express this a little more carefully in terms of the exterior of the image curve (Figure 2.2).

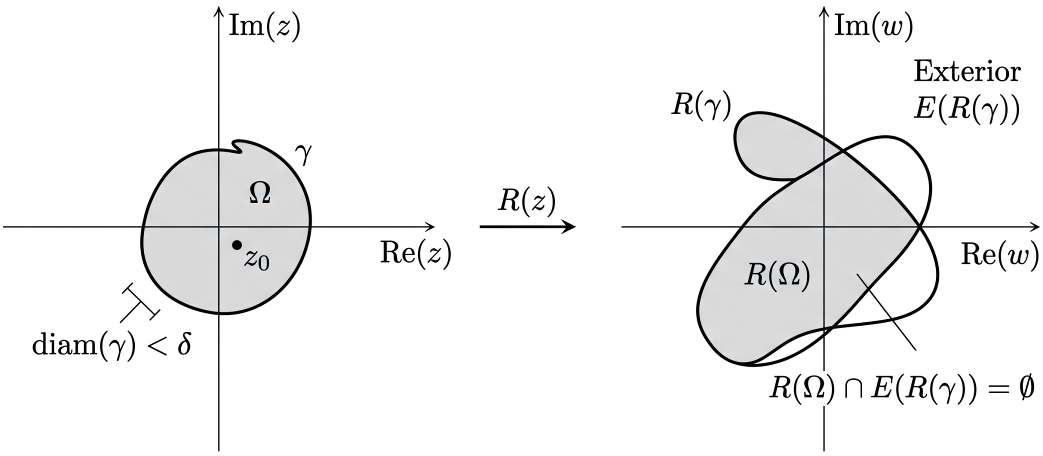


Figure 2.2: If the curve γ is sufficiently small ($\mathrm{diam}(\gamma) < \delta$), the image of its interior component Ω under a rational function R is entirely contained within the bounded regions defined by the image curve $\mathsf{R}(\gamma)$, never entering the exterior $\mathsf{E}(\mathsf{R}(\gamma))$.

2.4 Conjugacy

Definition 2.14 (Conjugate rational maps)**.** Given two rational maps R and S, we say R and S are *conjugate* if there is some Möbius map g with

$$\mathsf{S} = \mathsf{g}\mathsf{R}\mathsf{g}^{-1}.$$

Remark 2.16 (Conjugacy is an equivalence relation). Conjugacy is an equivalence relation, and the equivalence classes are *conjugacy classes* of rational maps.

Remark 2.17 (Invariant properties of conjugacy). Given two conjugate rational maps R and S, where $S = gRg^{-1}$. Then, the following are true:

1. $\deg(R) = \deg(S)$;
2. $S^n = gR^n g^{-1}$ (this means that we can transfer a problem concerning R to a (possibly simpler) problem concerning a conjugate S);
3. and S fixes $g(z)$ if and only if R fixes z (conjugacy respects fixed points).

All of these observations are elementary; nevertheless, they are important because, in general, we shall not bother to distinguish between conjugate rational functions.

As a simple illustration of the use of conjugacy, let us characterize the polynomials within the class of rational maps.

Remark 2.18 (Polynomials have one pole and it's ∞). A rational map R is a polynomial if and only if R has a pole at ∞ and no poles in $\mathbb{C}$; more succinctly, if and only if $R^{-1}\{\infty\} = \{\infty\}$.

Theorem 2.19 (Characterization of the conjugacy class of the polynomials)**.** *A non-constant rational map R is conjugate to a polynomial if and only if there is some w in $\mathbb{C}_\infty$ with $R^{-1}\{w\} = \{w\}$.*

2.5 Valency

Definition 2.15 (Valency of f at $z_0 \in \mathbb{C}$)**.** Given any function f that is non-constant and holomorphic near the point z_0 in $\mathbb{C}$. Then f has a Taylor expansion at z_0, say

$$f(z) = a_0 + a_k(z - z_0)^k + a_{k+1}(z - z_0)^{k+1} + \dots$$

where $a_k \neq 0$ and the positive integer k is uniquely determined by the condition that the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{(z - z_0)^k}$$

exists, is finite, and is non-zero.

We denote this integer k by $v_f(z_0)$ and call it the *valency*, or *order*, of f at z_0: of course, it is the number of solutions of $f(z) = f(z_0)$ at z_0.

Proposition 2.20 (Chain Rule for Valency)**.** *The valency function* ν *satisfies the important* Chain Rule*:*

$$\nu_{fg}(z_0) = \nu_f(g(z_0))\nu_g(z_0),$$

where f *and* g *are non-constant holomorphic functions near* $g(z_0)$ *and* z_0 *respectively; and* z_0*,* $g(z_0)$*, and* $fg(z_0)$ *are all in* $\mathbb{C}$*.*

Proof. See [1, pp. 37–38]. □

Remark 2.21 (Injectivity in terms of valency). Given any function f that is non-constant and holomorphic near the point z_0 in $\mathbb{C}$. Observe that f is injective in some neighborhood of z_0 if and only if $\nu_f(z_0) = 1$.

Remark 2.22 (Composing with injective functions preserves valency). Because of the previous remark (2.21), the Chain Rule shows that the valency is preserved under a pre-application, and a post-application, of an injective function.

Explicitly, if fg is defined near z with g injective near z, and if hf is defined near z with h injective near $f(z)$, then:

$$\nu_{fg}(z) = \nu_f(g(z)),$$
$$\nu_{hf}(z) = \nu_f(z).$$

The properties of valency under injective functions enable us to extend our definition of $\nu_f(z_0)$ to the case where $z_0 = \infty$, or $f(z_0) = \infty$ (or both).

Definition 2.16 (Valency at ∞)**.** Given any function f that is non-constant and holomorphic near the point ∞ in $\mathbb{C}_\infty$ or z_0 in $\mathbb{C}_\infty$ such that $f(z_0) = \infty$ (or both). Then, we can select any Möbius maps g and h which map z_0 and $f(z_0)$ respectively to points in $\mathbb{C}$ and then define $\nu_f(z_0)$ by

$$\nu_f(z_0) = \nu_F(g(z_0)),$$

where $F = hfg^{-1}$.

The essential observation is that if we repeat this procedure using different maps g and h, then the resulting answer will be the same; thus $\nu_f(z_0)$ is defined independently of the choice of g and h.

Definition 2.17 (Univalent map)**.** An analytic map $f : D \to \mathbb{C}_\infty$ is said to be *univalent* in D if it is injective there.

Remark 2.23 (f univalent $\implies \nu_f(z) = 1 \forall z \in D$). If f is univalent in D, then $\nu_f(z) = 1$ for every z in D.

The converse *need not hold*, being univalent is a global property (the whole domain), but valency is a local property (a neighborhood of z_0).

Example 2.1. The map $z \mapsto z^2$ of $\mathbb{C} - \{0\}$ onto itself has valency 1 everywhere, but it is not injective in $\mathbb{C} - \{0\}$.

Remark 2.24 (Rational maps are d-fold maps via valency). We've seen that rational maps are d-fold maps (Theorem 2.10), now we can write this in terms of valency. Given a non-constant rational map R of degree d, then for any z_0 in $R^{-1}\{w\}$, $\nu_R(z_0)$ is the number of solutions of $R(z) = w$ at z_0, and so for each w in the complex sphere,

$$\sum_{z \in R^{-1}\{w\}} \nu_R(z) = \deg(R).$$

2.6 The Inverse Function Theorem

Now that we have established the local behavior of analytic maps via valency, we can formalize when these maps can be locally inverted. In the classical complex plane, this is governed by the derivative.

Theorem 2.25 (Inverse Function Theorem in $\mathbb{C}$)**.** *Let $f : U \to \mathbb{C}$ be a holomorphic function on an open set $U \subseteq \mathbb{C}$, and let $z_0 \in U$. If $f'(z_0) \neq 0$, then there exist open neighborhoods $V \subseteq U$ of z_0 and W of $f(z_0)$ such that the restriction $f : V \to W$ is a bijection.*

Furthermore, the local inverse function $f^{-1} : W \to V$ is holomorphic, and its derivative satisfies:

$$(f^{-1})'(f(z_0)) = \frac{1}{f'(z_0)}.$$

Let us now bridge this classical theorem with our structural definition of valency at a point.

Remark 2.26 (Local injectivity in terms of derivative and valency in $\mathbb{C}$). By examining the Taylor expansion of a non-constant holomorphic function f near $z_0 \in \mathbb{C}$:

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \cdots$$

we see that $f'(z_0) \neq 0$ if and only if the first non-zero coefficient appears at index $k = 1$. By Definition 2.15, this means $\nu_f(z_0) = 1$.

Thus, the Inverse Function Theorem in $\mathbb{C}$ (Theorem 2.25) states that f possesses a local holomorphic inverse near $z_0 \in \mathbb{C}$ if and only if $\nu_f(z_0) = 1$.

Just as we did with valency, we can exploit the fact that composing with injective functions preserves local bijectivity to extend this theorem to cases where $z_0 = \infty$, $f(z_0) = \infty$, or both.

Theorem 2.27 (Inverse Function Theorem in $\mathbb{C}_\infty$)**.** *Let* $f : D_1 \to D_2$ *be a non-constant analytic map between two subdomains of* $\mathbb{C}_\infty$*, and let* $z_0 \in D_1$*. Then, we can select any two Möbius transformations* g *and* h *which map* z_0 *and* $f(z_0)$ *respectively to points in* $\mathbb{C}$*. Let* $F = hfg^{-1}$ *be the localized map defined near* $g(z_0) \in \mathbb{C}$*. If* $F'(g(z_0)) \neq 0$*, then* f *is a local bijection from a σ-neighborhood of* z_0 *onto a σ-neighborhood of* $f(z_0)$*. Consequently, it possesses a local analytic inverse* f^{-1} *near* $f(z_0)$*.*

The essential observation is that if we repeat this procedure using different maps g and h, then the resulting answer will be the same; thus the theorem can be generalized to $\mathbb{C}_\infty$ from $\mathbb{C}$ independently of the choice of g and h.

Remark 2.28 (Local injectivity in terms of localized derivative and valency in $\mathbb{C}_\infty$). Let f, g, h, and F be defined as in Theorem 2.27. By expanding the non-constant holomorphic map F into a Taylor series near the point $w_0 = g(z_0) \in \mathbb{C}$:

$$F(w) = F(w_0) + F'(w_0)(w - w_0) + \frac{F''(w_0)}{2!}(w - w_0)^2 + \cdots$$

we observe that $F'(g(z_0)) \neq 0$ if and only if the first non-zero coefficient after the constant term appears at index $k = 1$.

By your definition of extended valency, this means $\nu_F(g(z_0)) = \nu_f(z_0) = 1$. Thus, the Inverse Function Theorem in $\mathbb{C}_\infty$ (Theorem 2.27) establishes a universal rule across the entire Riemann sphere: an analytic map f admits a local analytic inverse near a point $z_0 \in \mathbb{C}_\infty$ if and only if $\nu_f(z_0) = 1$.

2.7 Fixed Points

Definition 2.18 (Fixed point of f)**.** Given any function $f : D \to \mathbb{C}_\infty$ defined on some domain $D \subseteq \mathbb{C}_\infty$. A point z_0 in D is a *fixed point* of f if $f(z_0) = z_0$.

Proposition 2.29 (Fixed points of rational maps)**.** *Given by two coprime polynomials* P *and* Q *and the rational map* $R = P/Q$*. Then,*

1. R *fixes* ∞ *if and only if* $\deg(P) > \deg(Q)$*;*

2. *and* R *fixes* $\zeta \in \mathbb{C}$ *if and only if* $P(\zeta) = \zeta Q(\zeta)$*; that is, the* fixed points of R in $\mathbb{C}$ *are the solutions of*

$$P(z) - zQ(z) = 0.$$

Proof. See [1, p. 39]. □

We count fixed points the same way we count zeros, that is, counting multiplicities.

Definition 2.19 (Number of fixed points at $z_0 \in \mathbb{C}$)**.** Given any function $f : D \longrightarrow \mathbb{C}_\infty$ defined on some domain $D \subseteq \mathbb{C}_\infty$. The number of fixed points of f at a point z_0 in $\mathbb{C}$ is given by the valency of the function $f(z) - z$ at z_0, i.e., $\nu_{f(z)-z}(z_0)$.

Example 2.2. The map $f(z) = z + z^3$ has three fixed points at 0. This is because the fixed points of f are the zeros of $f(z) - z$, and

$$f(z) - z = z^3,$$

which has valency 3 at 0 ($z^3 = f(0) - 0 = 0$ has three solutions). This is not to be confused with the valency of f at 0, which is 1. In fact, $f(z) = f(0)$ has only one solution at 0 ($f(z) = z(1 + z^2)$).

To count the number of fixed points at ∞, we need the following lemma.

Lemma 2.30. *Let* ζ *in* $\mathbb{C}$ *be a fixed point of an analytic map* f*, and let* φ *be any map that is analytic, injective and finite in some neighborhood of* ζ*. Then,* $\varphi f \varphi^{-1}$ *has the same number of fixed points at* $\varphi(\zeta)$ *as* f *has at* ζ*.*

Proof. See [1, pp. 39–40]. □

Definition 2.20 (Number of fixed points at ∞)**.** Given any function $f : D \longrightarrow \mathbb{C}_\infty$ defined on some domain $D \subseteq \mathbb{C}_\infty$, $\infty \in D$. Then we can select an analytic map φ, that is injective and finite in some neighborhood of ∞, and then define the number of fixed points of f at ∞ by the number of fixed points of $\varphi f \varphi^{-1}$ at $\varphi(\infty)$.

The essential observation is that if we repeat this procedure using different maps φ, then the resulting answer will be the same; thus, the number of fixed points of f at ∞ is defined independently of the choice of φ.

Moreover, as a direct consequence of the previous lemma, we have the following theorem.

Theorem 2.31 (Number of fixed points is invariant under conjugacy)**.** *Given two conjugate rational maps R and S, where $S = gRg^{-1}$, g is a Möbius transformation. Then, the number of fixed points of R at ζ is the same as the number of fixed points of S at $g(\zeta)$.*

Theorem 2.32 (Number of fixed points of a rational map)**.** *Given a rational map R of degree $d \geqslant 1$. Then, the number of fixed points of R in $\mathbb{C}_\infty$ is exactly $d + 1$ (counting multiplicities).*

Proof. See [1, p. 40]. □

Definition 2.21 (Multiplier of a fixed point in $\mathbb{C}$)**.** Given a rational map R and a fixed point ζ in $\mathbb{C}$ of R. We define *multiplier* of R at ζ, denoted by $m(R, \zeta)$, as the complex number given by the derivative:

$$m(R, \zeta) = R'(\zeta).$$

Definition 2.22 (Multiplier of the fixed point ∞)**.** Given a rational map R with a fixed point at ∞. Then, we can select a Möbius map g that maps ∞ to some point in $\mathbb{C}$, and then define the multiplier of R at ∞ by the multiplier of gRg^{-1} at $g(\infty)$:

$$m(R, \infty) = m(gRg^{-1}, g(\infty)).$$

Note that, by the preceding remarks, this definition is independent of the choice of g.

Example 2.3 (Multiplier at ∞ for a rational map)**.** Let R be a rational map of the form

$$R(z) = \frac{a_0 + a_1 z + \cdots + a_n z^n}{b_0 + b_1 z + \cdots + b_m z^m},$$

where $a_n b_m \neq 0$ and $n > m$, so R fixes ∞. By definition, $m(R, \infty)$ is the derivative of the conjugate map $S(z) = 1/R(1/z)$ at the origin. A direct computation shows:

$$S'(0) = \begin{cases} b_m/a_n & \text{if } n = m + 1, \\ 0 & \text{if } n > m + 1. \end{cases}$$

On the other hand, computing the derivative $R'(z)$ and taking the limit gives:

$$R'(\infty) = \lim_{z \to \infty} R'(z) = \begin{cases} a_n/b_m & \text{if } n = m + 1, \\ \infty & \text{if } n > m + 1. \end{cases}$$

Remark 2.33 (Formula for the multiplier of a rational map at a fixed point in $\mathbb{C}_\infty$). These results show that the multiplier $m(R, \zeta)$ of a rational map R at fixed point ζ is given by

$$m(R, \zeta) = \begin{cases} R'(\zeta) & \text{if } \zeta \neq \infty; \\ \frac{1}{R'(\infty)} & \text{if } \zeta = \infty. \end{cases}$$

Example 2.4. For example, if $R(z) = 3z$, then ∞ is an attracting fixed point of R (this is geometrically clear) and in this case, $m(R, \infty) = 1/3$.

We conclude this section by analyzing the behavior of the number of fixed points under iteration near a fixed point of f.

Theorem 2.34 (Iterates near the fixed point 0)**.** *Given an analytic function* f *that in a neighborhood of the origin has the expansion*

$$f(z) = az + b_1 z^{r+1} + \cdots ,$$

where $a \neq 0$, $b_1 \neq 0$, *and* $r \geqslant 1$. *Then its* n*-th iterate, in that neighborhood, has the form*

$$f^n(z) = a^n z + b_n z^{r+1} + \cdots ,$$

where

$$b_n = a^{n-1} b_1 \left(1 + a^r + a^{2r} + \cdots + a^{(n-1)r}\right).$$

Proof. See [1, p. 42]. □

From this theorem, we immediately obtain several important corollaries regarding the behavior of coefficients and the multiplicity of fixed points under iteration.

Corollary 2.34.1 ($a = 1 \implies b_n = nb_1$)**.** *If* $a = 1$, *then* $b_n = nb_1$.

Proof. This is a direct consequence of the formula for b_n:

$$b_n = a^{n-1} b_1 \left(1 + a^r + a^{2r} + \cdots + a^{(n-1)r}\right) = b_1(1 + 1 + \cdots + 1) = nb_1.$$

□

Corollary 2.34.2 ($b_n = 0 \iff a^r \neq 1$ and $a^{nr} = 1$)**.** $b_n = 0$ *if and only if* $a^r \neq 1$ *and* $a^{nr} = 1$.

Proof. The terms inside the parentheses in the formula for b_n form a geometric series. Thus, we can write:

$$b_n = a^{n-1} b_1 \frac{1 - a^{nr}}{1 - a^r}.$$

For b_n to be zero, the numerator of that fraction must be zero, but the denominator must not be zero (to avoid a 0/0 situation, which we already showed in the previous corollary (Corollary 2.34.1) that it evaluates to n).

$$\begin{aligned} 1 - a^{nr} &= 0 \implies a^{nr} = 1, \\ 1 - a^r &\neq 0 \implies a^r \neq 1. \end{aligned}$$

Thus, $b_n = 0$ if and only if $a^r \neq 1$ and $a^{nr} = 1$. □

Remark 2.35 (f^n has a single fixed point at the origin $\implies$ f has a single fixed point at the origin). If $a^n \neq 1$, then $a \neq 1$. Thus, if f^n has a single fixed point at the origin, then so does f.

Proof. If $a = 1$, then $a^n = 1$ for all n. Thus, if $a^n \neq 1$, then $a \neq 1$. Hence, if $a^n \neq 1$, then $a \neq 1$.

If $a \neq 1$, the function $f(z) - z$ looks like $(a-1)z + b_1 z^{r+1} + \cdots = z((a-1) + b_1 z^r + \cdots)$, where $a - 1 \neq 0$. Thus, $f(z)$ has a single fixed point at the origin.

If f^n has a single fixed point at the origin, then we can only factor out one z from $f^n(z) - z$, giving us $f^n(z) - z = (a^n - 1)z + b_n z^{r+1} + \cdots = z((a^n - 1) + b_n z^r + \cdots)$, where $a^n - 1 \neq 0$.

Combining these three results, we have that if f^n has a single fixed point at the origin, then $a^n \neq 1$, which implies $a \neq 1$, thus f has a single fixed point at the origin. □

Corollary 2.35.1 (Number of fixed points at 0 of the iterates)**.** *The iterate f^n has at least as many fixed points at the origin as f has. Furthermore, if f^n has strictly more fixed points at the origin than f, then $a \neq 1$ but $a^n = 1$.*

Proof. Let's begin by showing that f^n has at least as many fixed points at the origin as f has. We have two cases:

- If $a \neq 1$: $f(z) - z$ has a single fixed point at the origin. Since $f^n(0) = 0$, 0 must be a fixed point of f^n of multiplicity at least 1.
- If $a = 1$: then $f(z) - z = b_1 z^{r+1} + \cdots$, so 0 is a fixed point of f of multiplicity $r + 1$ ($b_1 \neq 0$). On the other hand, by Corollary 2.34.1, $f^n(z) - z = n b_1 z^{r+1} + \cdots$, so 0 is a fixed point of f^n of multiplicity $r + 1$ (the same as for f). Thus, if $a = 1$, then f^n has as many fixed points at the origin as f has.

Now, let's show that if f^n has strictly more fixed points at the origin than f, then $a \neq 1$ but $a^n = 1$. From our analysis above, if $a = 1$, the multiplicities are exactly equal, so for f^n to have strictly more fixed points at the origin than f, we cannot have $a = 1$. Thus $a \neq 1$.

If $a \neq 1$, then f has a single fixed point at the origin. For f^n to have strictly more fixed points at the origin than f, the term $(a^n - 1)$ from $f^n(z) - z = (a^n - 1)z + b_n z^{r+1} + \cdots = z((a^n - 1) + b_n z^r + \cdots)$ must vanish so that a higher power of z takes over. Thus, we must have $a^n = 1$. $\square$

2.8 Critical Points

Definition 2.23 (Critical point of a rational map)**.** Given a rational map R. A point z_0 in $\mathbb{C}_\infty$ is a *critical point* of R if R fails to be injective in any neighborhood of z_0, and if R is not constant. These points are precisely the points at which $v_R(z_0) > 1$.

Definition 2.24 (Critical value of a rational map)**.** Given a rational map R. A point w_0 in $\mathbb{C}_\infty$ is a *critical value* of R if it is the image of some critical point; that is, if $w_0 = R(z_0)$ for some critical point z_0 of R.

Remark 2.36. Given a rational function R of degree $d \geqslant 1$, and w a non-critical value of R. Then, $R^{-1}\{w\}$ consists of precisely d distinct points, say $z_1, z_2, \ldots, z_d$. As none of the z_j are critical points, there are neighborhoods of N of w, and $N_1, \ldots, N_d$ of $z_1, \ldots, z_d$ respectively, with R acting as a bijection from each N_j onto N. See Figure 2.3. It follows that for each j, the restriction R_j of R to N_j ($R : N_j \to N$) has an inverse

$$R_j^{-1} : N \to N_j.$$

Definition 2.25 (Branch of R^{-1} at a non-critical value w)**.** Given a rational function R of degree $d \geqslant 1$, and w a non-critical value of R. Then, we call each of the inverse maps $R_j^{-1} : N \to N_j$ a *branch* of R^{-1} at w.

By applying the Inverse Function Theorem (Theorem 2.25), we know that R is injective on some neighborhood of any point in $\mathbb{C}$ at which R' has neither a zero nor a pole: thus for all but a finite set of z, $v_R(z) = 1$ (recall that locally injective is the same as having valency 1 (Remark 2.21)) and, consequently,

$$\sum_{z \in \mathbb{C}_\infty} [v_R(z) - 1] < \infty.$$

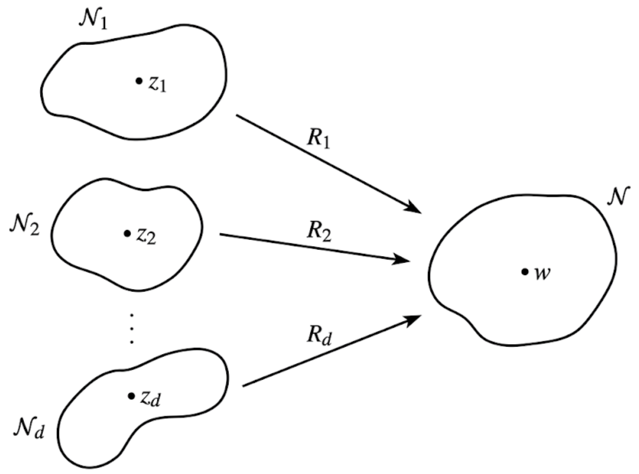


Figure 2.3: Bijective restrictions of R for a non-critical value w.

This sum gives us a measure of the number of multiple roots of R (and the difficulties in defining R^{-1}), and its actual value is given by the *Riemann-Hurwitz relation*.

Theorem 2.37 (Riemann-Hurwitz relation)**.** *Given a non-constant rational map* R*. Then,*

$$\sum_{z \in \mathbb{C}_\infty} [\nu_R(z) - 1] = 2d - 2.$$

Proof. See [1, p. 44]. □

The general term in the sum is positive only when z is a critical point of R so, as the valency is an integer, this provides us with an estimate of the number of critical points of R. For a non-constant polynomial P, we have $\nu_P(\infty) = \deg(P)$, so, in the same way, we obtain an estimate of the number of critical points of P in $\mathbb{C}$. These two estimates are given in the following corollary.

Corollary 2.37.1 (Bound on amount of critical points of a rational)**.** *A rational map of positive degree* d *has at most* $2d - 2$ *critical points in* $\mathbb{C}_\infty$*. A polynomial of positive degree* d *has at most* $d - 1$ *critical points in* $\mathbb{C}$*.*

Proof. Given a non-constant rational map R of degree d, for a point to be critical we must have $\nu_R(z) > 1$, so $\nu_R(z) - 1 \geqslant 1$. Thus, the number of critical points of R in $\mathbb{C}_\infty$ is at most $2d - 2$.

In the case of polynomials, we have $\nu_P(\infty) = \deg(P) = d$, so the contribution of ∞ to the sum in the Riemann-Hurwitz relation is $d - 1$. Thus, the number of critical points of P in $\mathbb{C}$ is at most $(2d - 2) - (d - 1) = d - 1$. □

We end this section with a simple result which locates the critical values of the iterate R^n in terms of the critical points of R.

Theorem 2.38 (Critical values of R^n). *Given a rational map R and its set of critical points C. Then, the set of critical values of R^n is*

$$R(C) \cup \cdots \cup R^n(C).$$

Proof. See [1, p. 45]. □

2.9 A Topology on the Rational Functions

To study the global dynamics and stability of families of rational maps, we must equip the set of rational functions with a robust topological structure.

Definition 2.26 (Spaces $\mathcal{C}$ and $\mathcal{R}$). Let $\mathcal{C}$ denote the class of all continuous maps from the extended complex plane into itself:

$$\mathcal{C} = \{f : \mathbb{C}_\infty \longrightarrow \mathbb{C}_\infty \mid f \text{ is continuous}\}.$$

We denote by $\mathcal{R}$ the specific subclass consisting of all non-constant and constant rational functions.

Definition 2.27 (Uniform Metric ρ). The *metric of uniform convergence* $\rho : \mathcal{C} \times \mathcal{C} \longrightarrow \mathbb{R}^+ \cup \{0\}$ on the complex sphere is defined by:

$$\rho(f, g) = \sup_{z \in \mathbb{C}_\infty} \sigma_0(f(z), g(z)),$$

where σ_0 is the spherical metric.

Since the chordal metric σ and the spherical metric σ_0 only differ by a bounded factor, we could equally well use the chordal metric here (the metric would change, but the topology would not).

Now $\mathcal{R}$ is a closed subset of $\mathcal{C}$ because if the rational functions R_n converge uniformly to R on the complex sphere, then R is analytic on the sphere and so it too is rational. More generally (and again we could use either the chordal metric or the spherical metric), we have the following theorem.

Theorem 2.39 (Uniform Limits Preserve Analyticity). *Suppose that each function in a sequence $\{f_n\}$ is analytic in a domain $D \subseteq \mathbb{C}_\infty$, and that f_n converges uniformly on D to f with respect to the spherical metric σ_0. Then f is also analytic in D.*

Proof. See [1, p. 46]. □

Corollary 2.39.1 ($\mathcal{R}$ is Closed)**.** *The space of rational functions $\mathcal{R}$ is a closed subset of the continuous function space C under the topology induced by the uniform metric ρ.*

With the topology established, we now investigate the topological behavior of the degree function. Crucially, the degree of a rational map cannot change abruptly under small uniform perturbations.

Theorem 2.40 (Continuity of the Degree Function)**.** *The degree map* $\deg : \mathcal{R} \longrightarrow \{0, 1, 2, \dots\}$ *is continuous. In particualr, if a sequence of rational functions R_n converges uniformly on the Riemann sphere to a function R, then R is rational and* $\deg(R_n) = \deg(R)$ *for all sufficiently large n.*

Proof. See [1, p. 46]. □

Definition 2.28 (Space of Degree n Maps)**.** We denote by $\mathcal{R}_n$ the collection of all rational maps of degree exactly equal to n:

$$\mathcal{R}_n = \{R \in \mathcal{R} \mid \deg(R) = n\}.$$

As a consequence of this theorem (Theorem 2.40), the class $\mathcal{R}_n$ of rational maps of degree n is an open subset of $\mathcal{R}$ for it is the inverse image of the open subset $\{n\}$ of $\mathbb{Z}$ under the continuous map deg. It is easy to see that each $\mathcal{R}_n$ is connected (for it R and S lie in $\mathcal{R}_n$, we can simply "slide" the zeros and poles of R to those of S while maintaining the same degree) so we obtain the following result.

Corollary 2.40.1 (Connected Components of $\mathcal{R}$)**.** *The subspaces $\mathcal{R}_n$ are the components of $\mathcal{R}$.*

There are a few remarks that I've not included in the notes, because Beardon points out that they can be safely ignored, in [1, p. 47].

Chapter 3

The Fatou and Julia Sets

3.1 The Fatou and Julia Sets

Bibliography

[1] Alan F. Beardon. *Iteration of rational functions*. First Paperback Edition. Graduate Texts in Mathematics. Springer, 2000. ISBN: 0387951512; 9780387951515.